

Lulu Wilson

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EDUCATION

University of Virginia, School of Engineering and Applied Science

August 2023 – May 2027

BS in Computer Science; BA in Mathematics

- Relevant Coursework: Data Structures, Algorithms, Software Development Essentials, Computer Systems Architecture, Discrete Mathematics, Cybersecurity, Software Engineering, Probability, Linear Algebra, Differential Equations, Real Analysis, Complex Analysis, Abstract Algebra, Statistics

Cumulative GPA: 3.94

EXPERIENCE

Software Engineering Intern

January 2026 – May 2026

City of Charlottesville IT Department

Charlottesville, VA

- Designed and implemented a full-stack ASP.NET MVC application in C# to replace a manual email-based notification workflow, introducing structured database operations and significantly improving efficiency.
- Architected a secure, automated, token-based email verification and unsubscribe pipeline, reducing manual processing and strengthening reliability and data integrity.
- Developed a RESTful API layer to handle HTTP requests and direct database interactions, enforcing separation of concerns through MVC architecture and enabling scalable, maintainable system design.

Teaching Assistant

August 2025 – Present

Advanced Data Structures and Algorithms, Discrete Mathematics

Charlottesville, VA

- Supported courses with 200+ students by leading review sessions and clarifying complex topics such as graph theory and advanced algorithms, helping students translate theoretical logic into practical code.
- Graded homework, quizzes, and exams, providing detailed, personalized feedback that helped students identify conceptual gaps and master fundamental proofs and Boolean logic.
- Collaborated with faculty to design and review exam questions, ensuring appropriate difficulty and comprehensive coverage of core computational and mathematical principles.

PROJECTS

Hoo Wants Some (Heroku Deployed Web App)

Aug 2025 – Dec 2025

- Developed a full-stack Django web application (Python, JavaScript, Tailwind) in a 5-person team to create a marketplace platform for UVA students, implementing listings, messaging, moderation, and notification systems.
- Led quality assurance efforts as QA lead by designing test plans, identifying defects, and coordinating fixes, improving system reliability and reducing post-release issues.
- Implemented a global notification system using JavaScript and Django signals to surface application-wide updates and improve user awareness and engagement.

Wikipedia Path Finder

Dec 2025

- Developed a Python application to compute shortest paths between Wikipedia pages by modeling large-scale link data as a graph structure.
- Implemented Dijkstra's algorithm using rustworkx to efficiently process thousands of nodes and edges.

SKILLS

Programming Languages: Python, Java, C, SQL, C#, HTML, CSS, Solidity

Databases: Relational Databases, SQLite

Frameworks & Libraries: JUnit, Gradle, Hibernate, JPA, Django, JavaFX, Mockito, React

Tools & Platforms: Git, GitHub, Make, LaTeX

Concepts & Methodologies: Object-Oriented Programming (OOP), Unit Testing, Agile Development, Blockchain Technologies

Software & Utilities: Excel, PowerPoint, Word, Outlook, Slack, Teams, Discord

EXTRACURRICULAR INVOLVEMENT

Affiliations: Girls Who Code (Hackathon Webmaster), First Year Players (Assistant Director, Social & Alumni Chair)

Achievements: Deans List (All semesters)

Certifications: Lifeguard Instructor (Red Cross), Lifeguard (Red Cross), Aquatics Instructor (Scouting America)